

Grading on a Curve? An Analysis of NYC Restaurant Health Inspection Scores

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1 Introduction

New York City grades every restaurant in its five boroughs through the Department of Health and Mental Hygiene (DOHMH) inspection system. Each inspection assigns a numerical score based on health code violations found, where a lower score is better. A score of 13 or below earns an “A,” 14 to 27 earns a “B,” and 28 or above earns a “C.” Restaurants post their letter grade in the window, making the grading system unusually visible to the public.

In 2014, data journalist Ben Wellington published an analysis noting a suspicious spike in inspection scores precisely at 13, the last score that qualifies for an A grade [1]. His hypothesis was that inspectors exercise informal leniency near grade boundaries, perhaps because issuing a score of 13 instead of 14 spares a restaurant the reputational cost of a B grade. The NYC Health Department responded that “inspectors are not instructed to offer leniency, just to cite what they see.” Wellington’s data, however, covered only 2010 to 2014, and the question has not been revisited formally with more recent data.

Our study has three aims. First, we test whether the grade-boundary spike persists in data updated through 2026 and quantify its magnitude statistically. Second, we ask whether a restaurant’s borough and cuisine type explain meaningful variation in inspection scores, using a two-way blocked ANOVA. Third, we examine whether violation types are distributed uniformly across boroughs, addressing the question of whether different parts of the city face different kinds of food safety challenges.

As a secondary predictive exercise, we also evaluate

how well pre-inspection characteristics alone, such as borough, cuisine, and neighborhood income, can forecast whether a restaurant will earn an A. This tests whether the outcome is predictable from context or whether it depends primarily on the restaurant’s own practices.

2 Data

2.1 Sources

The primary dataset is the DOHMH New York City Restaurant Inspection Results file, downloaded from NYC OpenData on April 3, 2026. It contains 113,571 records covering inspections from 2007 through 2026. The data is structured with one row per violation per inspection, so a single inspection event appears multiple times if multiple violations were cited. Key fields include borough, cuisine type, inspection score, inspection date, violation code, and inspection type (initial cycle, re-inspection, reopening, etc.).

Violation codes alone are not self-explanatory, so we used a secondary NYC Health Department reference file that maps each code to a descriptive category, such as “Pest Control,” “Cold Holding,” or “Handwash / Toilet.” We also merged neighborhood-level American Community Survey estimates at the 2020 Neighborhood Tabulation Area (NTA) level, including median worker earnings (MdeWrKE) and total population (Pop_1E), to capture socioeconomic context.

2.2 Preparation

Because the raw data has one row per violation, we deduplicated to one row per inspection event when computing scores and temporal trends, tak-

ing the first score value within each restaurant-date pair. This avoids artificially upweighting inspections with many violations.

Staten Island was excluded throughout due to sparse cell counts in the ANOVA design. Inspections with missing scores were also removed. The cleaned dataset contains 101,840 records covering 19,065 unique restaurants across four boroughs.

Cuisine types were grouped into ten *hypercategories* to reduce dimensionality and improve interpretability: American, East Asian, South & Southeast Asian, Latin American, European, Mediterranean, Middle Eastern, African, Dish-Type, and Beverages & Sweets. For example, Chinese, Japanese, Korean, and related cuisines were collapsed into “East Asian.” The Dietary Style category (Vegan, Vegetarian, etc.) was also excluded from the ANOVA due to sparse borough-level cell counts.

Table 1: Dataset summary after cleaning.

Statistic	Value
Inspection records	101,840
Unique restaurants	19,065
Years covered	2010–2026
A-grade rate	38.7%
Mean score (SD)	25.5 (19.3)
Median score	21

3 Statistical Models

3.1 Bump Test

To assess whether the spike at score 13 is statistically meaningful, we compared observed inspection counts across scores 12 to 21 against an expected uniform (flat) distribution. The expected count per score was estimated as the mean count at scores 14 to 21, a range safely away from both the A/B and B/C cutoffs. A chi-square goodness-of-fit test was applied to observed versus expected counts in this range.

3.2 Temporal Trend

We restricted to initial-cycle inspections between 2015 and 2025 and aggregated to one observation per inspection event. We then computed the mean

score by year and estimated the trend using ordinary least squares regression of mean score on year.

3.3 Two-Way Blocked ANOVA

To assess whether borough and cuisine type explain variance in inspection scores, we fit a two-way ANOVA with interaction:

$$\text{Score}_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

where α_i is the borough effect and β_j is the hypercategory effect. To keep the design balanced and prevent very large sample sizes from producing trivially small p -values, we sampled $n = 30$ observations per hypercategory-borough cell.

3.4 Violation Type by Borough

To answer whether certain violation types are associated with particular boroughs, we built a contingency table of violation category by borough and applied Pearson’s chi-square test of independence.

3.5 Predictive Logistic Regression

We fit a logistic regression using only features available before an inspection: borough, cuisine hypercategory, neighborhood median earnings, neighborhood population, and inspection year. The outcome is a binary A-grade indicator (score ≤ 13). Predictors were standardized on the training set. We used an 80/20 stratified train-test split and evaluated performance using AUC and accuracy.

4 Results

4.1 The Grade-Boundary Spike Is Large and Ongoing

Figure 1 shows the full inspection score distribution. Two spikes are immediately visible: one at score 13 (the A/B cutoff) and a smaller one at score 28 (the B/C cutoff). Both are consistent with boundary leniency.

The chi-square test is conclusive. Observed inspections at scores 12–13 total 17,863, against an expected 3,033 under a flat distribution. This is a surplus of 14,830 inspections, or +489% above

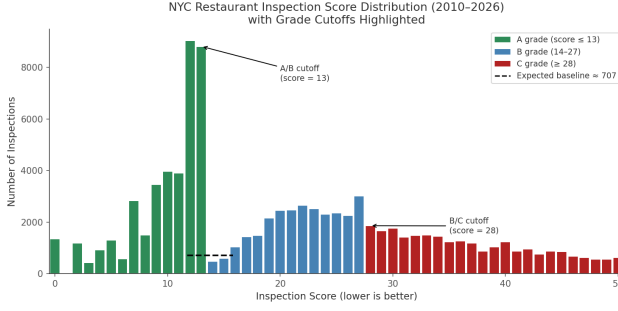


Figure 1: Inspection score distribution (2010–2026). Green bars are A-grade inspections; blue are B; red are C. The dashed line shows the expected flat baseline near the A/B cutoff.

expectation ($\chi^2 = 30,792$, $p \approx 0$). Wellington’s 2014 finding has not faded. Figure 2 provides a zoomed view centered on the A/B boundary.

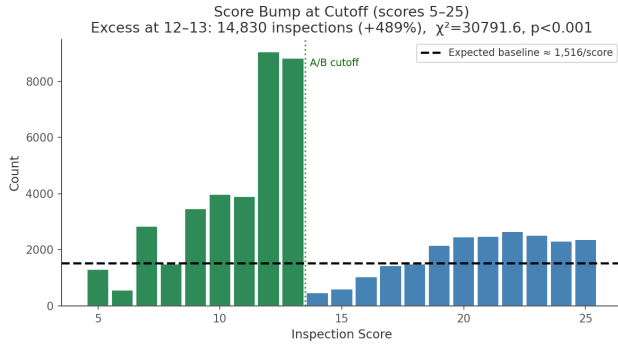


Figure 2: Zoomed view of scores 5–25. The dashed line shows the expected flat baseline. The spike at 12–13 represents roughly 15,000 inspections above what would be expected under uniform grading.

4.2 Inspection Scores Have Worsened Substantially

Figure 3 shows mean initial-cycle inspection scores from 2015 to 2025, after deduplicating to one observation per inspection event. The mean score increased from 11.3 in 2015 to 19.0 in 2025. A simple linear regression estimates the rate of increase at 0.87 points per year ($R^2 = 0.91$, $p < 0.001$). At this pace, the average restaurant has shifted from comfortably earning an A on an initial inspection in 2015 to scoring squarely in B-grade territory by 2025.

The causes of this trend are not clear from the data

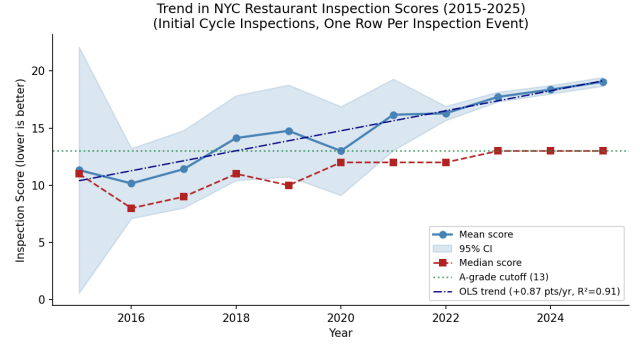


Figure 3: Mean and median inspection score by year (initial-cycle inspections only, one row per inspection event). The green dotted line marks the A-grade cutoff at 13.

alone. Possible explanations include genuine deterioration in restaurant hygiene, increased inspection stringency, changes in the restaurant population, or some combination. Disentangling these explanations would require inspector and restaurant fixed effects, which is left for future work.

4.3 Borough and Cuisine Have Modest but Significant Effects

Figure 4 shows score distributions by borough and by cuisine hypercategory. Manhattan restaurants score best (median 19), followed by the Bronx and Brooklyn (both median 20), with Queens scoring worst (median 22). These differences are present but modest in absolute terms.

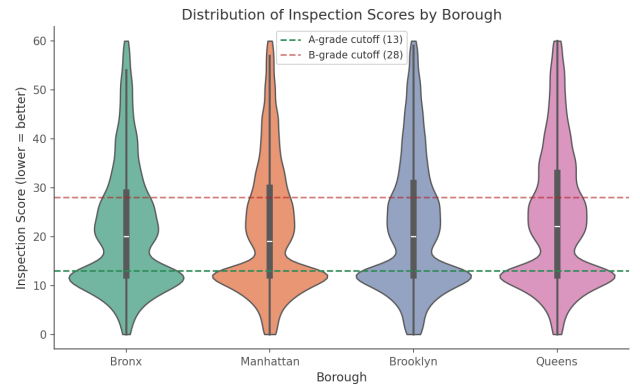


Figure 4: Score distributions by borough. Lower is better. Medians range from 19 (Manhattan) to 22 (Queens).

The two-way blocked ANOVA on $n = 30$ observa-

tions per cell confirms these patterns. Cuisine hypercategory is a significant predictor ($F(9, 1160) = 4.12$, $p < 0.001$), as is the hypercategory-by-borough interaction ($F(27, 1160) = 1.54$, $p = 0.038$). Borough alone is only marginally significant ($F(3, 1160) = 2.25$, $p = 0.081$). Effect sizes are modest: the total residual variance ($SS_{\text{res}} = 482,194$) dwarfs the variance attributable to borough and cuisine combined ($SS_{\text{model}} = 35,521$), suggesting that individual restaurant practices explain far more of the outcome than geography or cuisine type.

4.4 Violation Types Vary Significantly Across Boroughs

Figure 5 shows the rate of each violation category by borough, expressed as a percentage of all inspections in that borough. The chi-square test strongly rejects independence between violation type and borough ($\chi^2 = 678.2$, $df = 66$, $p \approx 2 \times 10^{-102}$).

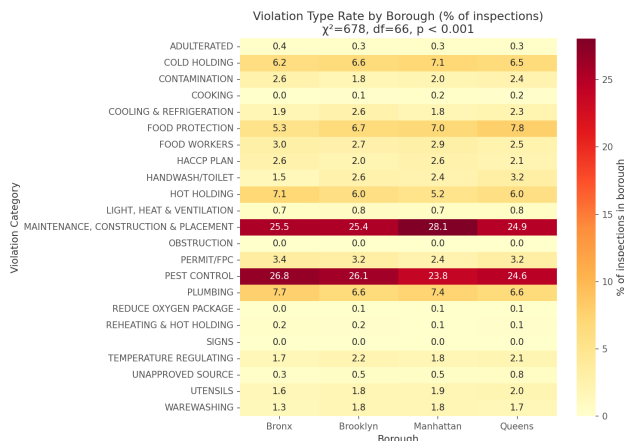


Figure 5: Violation category rates by borough (% of inspections). Darker cells indicate higher violation rates. Pest Control and temperature-related violations show the strongest geographic variation.

Pest Control violations are more concentrated in the Bronx relative to other boroughs. Cooling and Refrigeration violations are elevated in Manhattan. Queens and Brooklyn show higher rates of Hot Holding violations. These patterns likely reflect differences in restaurant type, building infrastructure, and neighborhood environment rather than inspection rigor alone, though the data cannot

distinguish between these explanations.

4.5 Pre-Inspection Features Cannot Reliably Predict Grade

The logistic regression using only pre-inspection features achieves an AUC of 0.60 and an accuracy of 59.4% on the held-out test set (Table 2). Both figures are barely above a baseline classifier. The ROC curve is shown in Figure 6.

Table 2: Predictive model performance.

Model	AUC	Accuracy
Pre-inspection features	0.60	59.4%

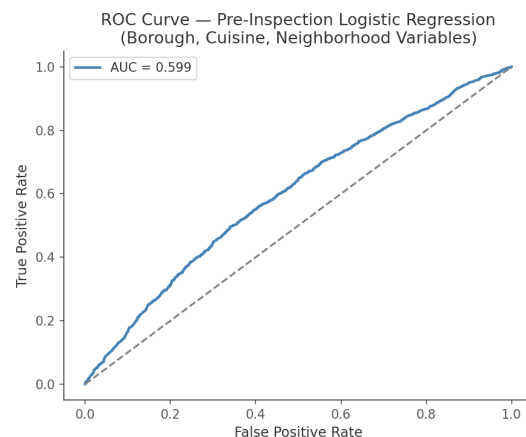


Figure 6: ROC curve for the pre-inspection logistic regression. An AUC near 0.60 indicates limited predictive power from borough, cuisine, and neighborhood variables alone.

This near-chance performance has a clear interpretation: knowing where a restaurant is located, what cuisine it serves, and how wealthy its neighborhood is tells us very little about whether it will pass inspection. Restaurant-level practices and management quality appear to drive outcomes far more than contextual factors.

5 Model Validation

The predictive logistic regression was evaluated on a stratified 80/20 hold-out split, with standardization fit only on training data. No information from the test set influenced model training. The modest

AUC of 0.60 reflects the genuine difficulty of predicting inspection outcomes from pre-inspection characteristics rather than any deficiency in the modeling approach.

For the ANOVA, we used balanced sampling across cells ($n = 30$ per cell) to avoid the problem common in large observational datasets where even negligible effect sizes produce highly significant p -values. The results should be interpreted with effect sizes in mind rather than p -values alone. Residual variance far exceeds the variance explained by the model terms, placing clear limits on how much the between-group differences matter practically.

The chi-square tests for the bump and the violation-by-borough analysis use the full dataset and are therefore extremely well-powered. Effect sizes in both cases are substantive: the 489% excess at the grade boundary is not a statistical artifact, and the geographic variation in violation types is pronounced enough to be visible in the raw heatmap.

6 Conclusion

Four findings stand out from this analysis.

Grade-boundary leniency is real and has not diminished since Wellington’s 2014 report. Nearly 15,000 inspections appear at scores 12–13 in excess of what a fair distribution would predict. The NYC Health Department’s denial notwithstanding, the pattern in the data is striking and consistent.

Inspection scores have worsened materially over the past decade. The mean initial-inspection score rose by roughly 0.87 points per year from 2015 to 2025, moving from A-grade territory to solidly B-grade territory. Whether this reflects actual deterioration in restaurant conditions or a shift in inspection behavior is an important open question.

Violation types are not geographically uniform. Different boroughs show distinct violation profiles, with Pest Control violations elevated in the Bronx and temperature-related violations more prevalent in Manhattan and outer boroughs. This geographic heterogeneity is highly statistically significant and may be useful for targeted public health outreach.

Finally, context does not predict outcomes. With

an AUC of 0.60, knowing a restaurant’s borough, cuisine, and neighborhood income cannot meaningfully forecast whether it will earn an A. This reinforces the conclusion from the ANOVA: most of the variation in inspection scores is within-group, driven by factors specific to individual restaurants rather than their broader context.

Future work could examine whether the grade-boundary spike varies by borough or inspector team, or test whether the temporal worsening trend persists after controlling for changes in the restaurant population.

Appendix A: Literature Review

Appendix B: ANOVA Table

Table 3: Two-way ANOVA results ($n = 30$ per cell, Bronx and African cuisine as baselines, Staten Island and Dietary Style excluded).

Term	Df	F	p	
Hypercategory	9	4.12	< 0.001	***
Borough	3	2.25	0.081	.
Hypercategory $\times$ Borough	27	1.54	0.038	*
Residuals	1160			

Signif.: *** $p < 0.001$, * $p < 0.05$, . $p < 0.1$

Appendix C: Additional Figures

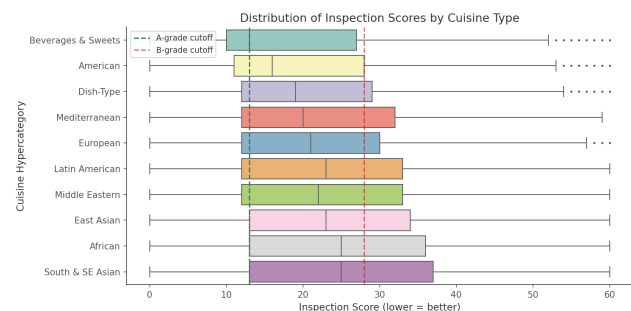


Figure 7: Score distributions by cuisine hypercategory. African and European restaurants score best on average; Dish-Type and Beverages & Sweets score worst.

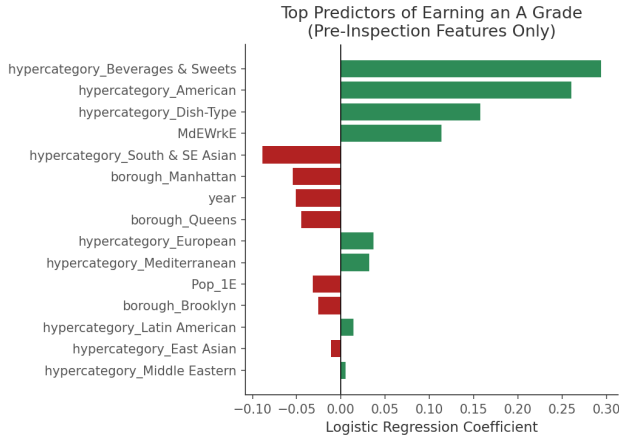


Figure 8: Top 15 coefficients from the pre-inspection logistic regression. Green indicates positive association with earning an A; red indicates negative association. No predictor has a large coefficient, consistent with the low AUC.

Appendix D: Code

Python analysis code is in `new_analysis.py`. The R analysis and ANOVA are in `group_analysis.Rmd`. Both files are submitted alongside this report.

References

- [1] Wellington, B. (2014). *Think NYC Restaurant Grading is Flawed? Here's the Proof*. I Quant NY. <https://iquantny.tumblr.com/post/76928412519>